

第五章 金属电子论

金属性质 $\sim$ 导电电子 $\sim$ Fermi 分布

一、Fermi 分布函数

$$f(E) = \frac{1}{e^{(E-\mu)/kT} + 1}$$

电子数守恒:

$$\int_0^{\infty} f(E) N(E) dE = N$$

二、基态 $T=0K$ 下的分布函数 $f_0(E)$ 和 Fermi 能 E_F

$$f_0(E) = \theta(E_F - E) = \begin{cases} 1 & E \leq E_F \\ 0 & E > E_F \end{cases}$$

$$E_F = \mu(0)$$

$$\frac{\partial f_0(E)}{\partial E} = \theta'(E_F - E) = \delta(E_F - E)$$

对 E 积分由电子 n :

$$E(\vec{k}) = \frac{\hbar^2}{2m} k^2 \quad N(E) = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2}$$

$$N = \int_0^{\infty} f_0(E) N(E) dE$$

$$= \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \int_0^{E_F} E^{1/2} dE = \frac{2}{3} \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} E_F^{3/2}$$

$$\text{解得 } E_F = \frac{\hbar^2}{2m} \left(\frac{3\pi^2 N}{V}\right)^{2/3} = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3} \sim n^{2/3}$$

$$n = \int_0^{E_F} E N(E) dE = \frac{V}{5\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} E_F^{5/2} = \frac{3}{5} N E_F$$

$$E(\vec{k}) = \frac{\hbar^2 k^2}{2m} = \frac{3}{5} E_F$$

零温压强

$$P = n - TS$$

$$P_0 = - \left(\frac{\partial F_0}{\partial V}\right)_T = - \frac{\partial U_0}{\partial V} = \frac{2}{3} \frac{U_0}{V} = \frac{2}{3} n E_F$$

三、激发态 $T \neq 0K$

$$f(E) = \frac{1}{e^{(E-\mu)/kT} + 1} = \begin{cases} 1 & E-\mu \ll kT \\ \frac{1}{2} & E-\mu = 0 \\ 0 & E-\mu \gg kT \end{cases}$$

$$\frac{\partial f(E)}{\partial E} = -\frac{1}{kT} \frac{e^{(E-\mu)/kT}}{(e^{(E-\mu)/kT} + 1)^2} = -\frac{1}{kT} \frac{1}{e^{(E-\mu)/kT} + 1} \cdot \frac{1}{e^{-(E-\mu)/kT} + 1}$$

利用归一化条件求出 $\mu(T)$

$$N = \int_0^\infty f(E) N(E) dE \quad Q(E) = \int_0^\infty N(E) dE$$

$$= \int_0^\infty f(E) dQ(E) = f(E) Q(E) \Big|_0^\infty - \int_0^\infty Q(E) df(E)$$

$$= - \int_0^\infty Q(E) \frac{\partial f(E)}{\partial E} dE$$

$$\frac{\partial f(E)}{\partial E} \sim \delta(E-\mu) \text{ 在 } E=\mu \text{ 附近展成}$$

$$= Q(\mu) \int_0^\infty -\frac{\partial f(E)}{\partial E} dE + Q'(\mu) \int_0^\infty (E-\mu) \left(-\frac{\partial f}{\partial E}\right) dE + \frac{1}{2} Q''(\mu) \int_0^\infty (E-\mu)^2 \left(-\frac{\partial f}{\partial E}\right) dE \quad \xi = \frac{E-\mu}{kT}, \int_0^\infty \rightarrow \int_{-\infty}^\infty$$

$$= Q(\mu) \cdot 1 + Q'(\mu) (kT) \int_{-\infty}^\infty \xi d\xi + \frac{1}{2} Q''(\mu) (kT)^2 \int_{-\infty}^\infty \xi^2 d\xi$$

$$+ \frac{1}{6} Q'''(\mu) (kT)^3 \int_{-\infty}^\infty \xi^3 d\xi$$

$$= Q(\mu) + \frac{1}{6} Q'''(\mu) (kT)^3 \frac{\pi^2}{3}$$

$$\approx Q(E_F^0) + (\mu - E_F^0) Q'(E_F^0) + \frac{\pi^2}{6} Q''(E_F^0) (kT)^2$$

$$= N + (\mu - E_F^0) Q'(E_F^0) + \frac{\pi^2}{6} Q''(E_F^0) (kT)^2 \quad Q(E_F^0) = \int_0^\infty N(E) dE = N$$

$$\Rightarrow \mu(T) = E_F^0 - \frac{\pi^2}{6} \frac{Q''(E_F^0)}{Q'(E_F^0)} (kT)^2 = E_F^0 - \frac{\pi^2}{12} \frac{1}{E_F^0} (kT)^2$$

$$= E_F^0 \left[1 - \frac{\pi^2}{12} \left(\frac{kT}{E_F^0} \right)^2 \right]$$

简并性, p_F^0, E_F^0 很大

则据, $\mu(T) \sim E_F \gg kT, T \ll T_F$

在室温下, 自由电子气高度简并的。

四、自由电子气的比热容

$$\text{电子气总能量 } U(T) = \int_0^\infty E N(E) f(E) dE \quad R = \int_0^E E N(E) dE$$

$$R' = E N(E)$$

$$= \int_0^\infty R'(E) f(E) dE$$

$$= R(E) f(E) \Big|_0^\infty - \int_0^\infty R(E) f'(E) dE$$

$$= - \int_0^\infty R(E) f'(E) dE$$

$$= - R(\mu) \int_0^\infty f'(E) dE - R'(\mu) \int_0^\infty (E-\mu) f'(E) dE - \frac{1}{2} R''(\mu) \int_0^\infty (E-\mu)^2 f'(E) dE$$

$$= - R(\mu) \int_0^\infty f'(E) dE - \frac{1}{2} R''(\mu) \int_0^\infty (E-\mu)^2 f'(E) dE$$

$$f(\infty) - f(0) = 1$$

$$= R(\mu) + \frac{1}{2} R''(\mu) (kT)^2 \frac{\pi^2}{6}$$

$$= R(E_F) + (\mu - E_F) R'(E_F) + \frac{\pi^2}{6} R''(E_F) (kT)^2$$

$$= U_0 - \frac{\pi^2}{12 E_F} (kT)^2 E_F N(E_F) + \frac{\pi^2}{6} [N(E_F) + E_F N'(E_F)] (kT)^2$$

$$= U_0(T) + \frac{\pi^2}{6} \underbrace{N(E_F)} (kT)^2$$

电子气的激发能量，

在 E_F 附近 kT 的电子数 $N(E_F) kT \times$ 平均热激发能 kT

$$\text{比热容 } C_V = \left(\frac{\partial U}{\partial T} \right)_V = \frac{\pi^2}{3} N(E_F) k^2 T \quad N(E_F) = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar} \right)^{3/2} E_F^{1/2}$$

$$= \frac{\pi^2}{3} \frac{3}{2} \frac{N}{E_F} k^2 T$$

$$= \frac{3}{2} \frac{N}{E_F}$$

$$= \frac{\pi^2}{2} N k \left(\frac{T}{T_F} \right) \sim T$$

室温下 $T \ll T_F$, $C_V \ll Nk$

$$\text{金属比热容 } C_V = C_{V\text{晶}} + C_{V\text{电}} = \gamma T + \beta T^3$$

§ 5.2 金属的费米面

在空间中 E_F 的等能面。

$$k = \left(\frac{2mE_F}{\hbar^2} \right)^{1/2} = (3\pi^2 n)^{1/3}$$

二、实际金属的费米面。

1. 碱金属

bcc 结构，每元胞 1 个价电子 - $n = \frac{2}{a^3}$ (当时不是 1)

$$k_F = 0.62 \left(\frac{2}{a} \right)$$

§ 0.3 费米面的实验测定

一、回旋共振。

$$\omega_c = \frac{eB}{m_e^*}$$

二、德哈斯-范阿尔芬效应。

§ 5.4 输运现象

$$\begin{cases} \vec{J}_n = -n \vec{v} T \\ \vec{J}_n = -D \nabla n \\ \vec{J}_e = -\sigma \nabla \varphi \end{cases}$$

一、非平衡分布函数

$$\vec{J}_e = -\frac{ze}{(2\pi)^3} \int \vec{v}(\vec{k}) f(\vec{k}) d^3 \vec{k}$$

平衡态下, $T_n[E(\vec{k}), T] = e^{[E(\vec{k}) - \mu]/kT - 1}$

$$\varepsilon(\vec{k}) = \varepsilon(-\vec{k}) \rightarrow f(\vec{k}, T) = f(-\vec{k}, T)$$

$$\vec{v} = \frac{1}{\hbar} \nabla_{\vec{k}} E(\vec{k}) \rightarrow v(\vec{k}) = -v(-\vec{k})$$

$$\Rightarrow J_e|_n = 0 \quad \text{电流为零}$$

二、Boltzmann 方程

$$\frac{df}{dt} = \left(\frac{df}{dt} \right)_{\text{漂}} + \left(\frac{df}{dt} \right)_{\text{碰}}$$

1. 漂移

$$\begin{aligned} \left(\frac{df}{dt} \right)_{\text{漂}} &= -\vec{v}_k \cdot \left[f(\vec{k}, \vec{r}, t) \frac{d\vec{k}}{dt} \right] - \vec{v}_r \cdot \left[f(\vec{k}, \vec{r}, t) \frac{d\vec{r}}{dt} \right] \\ &= -\frac{d\vec{k}}{dt} \cdot \vec{v}_k f(\vec{k}, \vec{r}, t) - f(\vec{k}, \vec{r}, t) \vec{v}_k \cdot \frac{d\vec{k}}{dt} \\ &\quad - \frac{d\vec{r}}{dt} \cdot \vec{v}_r f(\vec{k}, \vec{r}, t) - f(\vec{k}, \vec{r}, t) \vec{v}_r \cdot \frac{d\vec{r}}{dt} \end{aligned}$$

$$\vec{v}_k \cdot \frac{d\vec{k}}{dt} = \frac{d}{dt} (\vec{v}_k \cdot \vec{k}) = 0, \quad \vec{v}_r \cdot \frac{d\vec{r}}{dt} = \frac{d}{dt} (\vec{v}_r \cdot \vec{r}) = 0$$

$$\left(\frac{df}{dt} \right)_{\text{漂}} = -\frac{d\vec{k}}{dt} \cdot \vec{v}_k f(\vec{k}, \vec{r}, t) - \frac{d\vec{r}}{dt} \cdot \vec{v}_r f(\vec{k}, \vec{r}, t)$$

2. 碰撞. (不可逆过程)

散射概率 $\vec{k} \rightarrow \vec{k}'$, $\Theta(\vec{k}, \vec{k}')$

单位时间 $\vec{k} \rightarrow \vec{k}'$

$\vec{k}'$ 态未被占据概率

$$a = \frac{1}{(2\pi)^3} \int \left[\gamma(\vec{k}, \vec{r}, t) \left(1 - \gamma(\vec{k}', \vec{r}, t) \right) \right] \Theta(\vec{k}, \vec{k}') d\vec{k}'$$

$$\vec{k}' \rightarrow \vec{k}$$

$$b = \frac{1}{(2\pi)^3} \int \left[\gamma(\vec{k}', \vec{r}, t) \left(1 - \gamma(\vec{k}, \vec{r}, t) \right) \right] \Theta(\vec{k}', \vec{k}) d\vec{k}'$$

$$\left\langle \frac{\partial f}{\partial t} \right\rangle_{\text{碰撞}} = b - a$$

对于稳态问题, $\frac{\partial f}{\partial t} = 0$

$$\frac{d\vec{k}}{dt} \cdot \nabla_{\vec{k}} f + \frac{d\vec{r}}{dt} \cdot \nabla_{\vec{r}} f = b - a$$

§ 5.5 金属的电导率

$$\left(\frac{\partial f}{\partial t}\right)_{\text{碰撞}} : \psi - \psi_0 = - \frac{\psi - \psi_0}{\tau(k)}$$

$$\begin{aligned} \text{Cu: } \tau &\sim 10^{-14} \text{ (T=4K)} \\ &\sim 10^{-17} \text{ (RT)} \end{aligned}$$

二. 电导率公式

$$- \frac{e}{\hbar} \vec{E} \cdot \nabla_k f(k) = \frac{f - f_0}{\tau(k)}$$

对 f

§ 5.6.

在平衡态下，

$$\theta(\vec{k}', \vec{k}) \left[p_0(\vec{k}') / (1 - p_0(\vec{k})) \right] = \theta(\vec{k}, \vec{k}') \left[p_0(\vec{k}) / (1 - p_0(\vec{k}')) \right]$$

$$\theta(\vec{k}', \vec{k}) = \frac{e^{\frac{1}{2}(E(\vec{k}) - \mu)/kT}}{e^{\frac{1}{2}(E(\vec{k}') - \mu)/kT} + 1} \frac{e^{\frac{1}{2}(E(\vec{k}) - \mu)/kT}}{e^{\frac{1}{2}(E(\vec{k}') - \mu)/kT} + 1}$$

$$= \theta(\vec{k}, \vec{k}') \frac{e^{\frac{1}{2}(E(\vec{k}) - \mu)/kT}}{e^{\frac{1}{2}(E(\vec{k}) - \mu)/kT} + 1} \frac{e^{\frac{1}{2}(E(\vec{k}') - \mu)/kT}}{e^{\frac{1}{2}(E(\vec{k}') - \mu)/kT} + 1}$$

$$\Rightarrow \theta(\vec{k}', \vec{k}) e^{\frac{E(\vec{k}) - E(\vec{k}')}{kT}} = \theta(\vec{k}, \vec{k}')$$

对于弹性散射 $E(\vec{k}) = E(\vec{k}')$, $\theta(\vec{k}, \vec{k}') = \theta(\vec{k}', \vec{k})$

§ 5.7

$$\rho = \frac{m^*}{n e} \frac{1}{2U_0} \quad \left\{ \right.$$

一、随时间变化的微扰势

$$T=0 \text{ 时, } U(\vec{r}) = \sum_i u(\vec{r} - \vec{R}_i), \quad \vec{R}_i = \sum_i l_i \vec{a}_i$$

$T \neq 0$ 时, 晶格振动 u_i 为微扰

$$\hat{H} = -\sum_i u_i \cdot \nabla u(\vec{r} - \vec{R}_i)$$

二、散射概率 $\theta(E, E')$

Fermi 黄金规则

$$\theta(E, E') = \frac{2\pi}{\hbar}$$

1. 能量守恒

$$E(E') = E(E) + \hbar\omega(\vec{q})$$

$$E(E') = E(E) - \hbar\omega(\vec{q})$$

2. 动量守恒 -

3. 正常心 / 倒逆 u 过程.

电子-电子相互作用:

$$\frac{1}{\epsilon(\vec{k})} \propto \int d\vec{r}, d\vec{r}', d\vec{r}'', d\vec{r}''' \rho_0(\vec{r}) [1 - \rho_0(\vec{r}')] \rho_0(\vec{r}'') [1 - \rho_0(\vec{r}''')] \\ (1 - \cos \eta) \delta_{\vec{r} + \vec{r}' - \vec{r}'' - \vec{r}'''} \delta_{\vec{k} + \vec{k}' - \vec{k}'' - \vec{k}'''} d\vec{k}' \propto \vec{r}'' d\vec{k}''$$

散射发生在费米面附近: $E \sim E' \sim E'' \sim E''' \sim E_F$

$$\rho_0(\vec{r}) [1 - \rho_0(\vec{r}')] \approx \rho_0(E_F) [1 - \rho_0(E_F)] = kT \left(\frac{\partial \rho_0}{\partial E} \right) \\ \sim \rho_0(\vec{r}'') [1 - \rho_0(\vec{r}''')] \\ \rho \sim \frac{1}{\epsilon(\vec{k})} \sim T^2$$

§ 3.8 等离子体

1. 为什么金属中的电子可以视为近自由？屏蔽

2. 金属体系对电磁波的响应。

一、等离子体

自由电子气：均匀正离子+电子， $\rho_0^+ = \rho_0^- = \rho_0$



电子受微扰后电荷密度 $\rho(\vec{r}, t)$

$$\begin{cases} \nabla \cdot \vec{E} = \frac{\rho - \rho_0}{\epsilon_0} \\ m \frac{d\vec{v}}{dt} = -e \vec{E} \\ \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \end{cases}$$

$$\rho \sim \rho_0 \quad \frac{\partial \rho}{\partial t} + \rho_0 \nabla \cdot \vec{v} = 0$$

$$\frac{\partial^2 \rho}{\partial t^2} + \rho_0 \nabla \cdot \frac{d\vec{v}}{dt} = \frac{\partial^2 \rho}{\partial t^2} + \rho_0 \nabla \cdot \left(-\frac{e}{m} \vec{E} \right)$$

$$= \frac{\partial^2 (\rho - \rho_0)}{\partial t^2} - \frac{e^2 \rho_0}{\epsilon_0 m} (\rho - \rho_0) = 0 \quad \rho_0 = ne$$

$$\Rightarrow \omega_p^2 = \frac{ne^2}{\epsilon_0 m}$$

二、电子气的个别激发。

基态：Fermi 球内全满，球外全空。

Fermi 球内电子 $\vec{k} \rightarrow$ 球外电子 $\vec{k} + \vec{q}$

$\hookrightarrow$ 空穴 $\vec{k}$ —— 为 $\vec{k}$ 所激发的 $\vec{k} + \vec{q}$

$$\hbar \omega_q = \frac{\hbar^2}{2m} \left[(\vec{k} + \vec{q})^2 - k^2 \right] = \frac{\hbar^2}{2m} (\vec{q}^2 + 2\vec{k} \cdot \vec{q})$$

$$|\vec{k}| < k_F < |\vec{k} + \vec{q}|$$

①
三、静电屏蔽，准电子。

电子 + 正电荷屏蔽云 = 准电子。

$q\delta(\vec{r})$

$$\begin{aligned}n(\vec{r}) &= \int \varphi(E - e\varphi(\vec{r})) N(E) dE \\&\approx \int \varphi(E - e\varphi(\vec{r}) \frac{\partial \varphi}{\partial E}) N(E) dE \\&= n + e\varphi(\vec{r}) \int \left(\frac{\partial \varphi}{\partial E} \right) N(E) dE \\&= n + e\varphi(\vec{r}) N(E_F) \quad \rightarrow \Delta(E - E_F) \\&\quad \text{= 0?}\end{aligned}$$

$$\rho(\vec{r}) = q\delta(\vec{r}) - en(\vec{r}) = q\delta(\vec{r}) - e^2\varphi(\vec{r})N(E_F)$$

满足 Poisson 方程：

$$\nabla^2 \varphi(\vec{r}) = -\frac{\rho(\vec{r})}{\epsilon_0} = -\frac{1}{\epsilon_0} (q\delta(\vec{r}) - e^2\varphi(\vec{r})N(E_F))$$

在体空间中求解

12. 等离子体中的横振荡.

$$\left\{ m \frac{d\vec{v}}{dt} = -e\vec{E} \right.$$